QP CODE: **24900140**

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR

EXAMINATION, NOVEMBER 2024

First Semester

Discipline Specific Core Course - MG1DSCHTC100 - BASIC HORTICULTURE

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A. Short Answers

Answer any ten questions. Each question carries 1 mark.

1. **What are the primary objectives of horticulture?**

   - Enhancing Crop Yield and Quality
   - Promoting Sustainable Practices.
   - Improving Livelihoods.
   - Aesthetic and Recreational Benefits.
   - Research and Development.
   - Supporting Public Health. (**Any suitable point**)

2. **How does horticulture aim to improve water use efficiency and soil health in plant cultivation?**

   Horticulture improves water use efficiency by employing techniques like drip irrigation and mulching, and it enhances soil health by incorporating organic matter and adopting crop rotation practices. **(Liberal valuation)**

3. **What is a vertical garden?**

   A method of growing plants on vertically suspended panels, using various means.

4. **Differentiate living and non-living components of a garden.**

   - Living components mainly plants and other organisms. Eg: trees, shrubs, shrub, grass etc.
   - The non-living elements/ garden adornments artificial structures in the garden. Eg: Bunces, rocks, statues, fountains etc.

5. **What does the term water-use-efficiency refer to in irrigation?**

   It is the ratio of the amount of water effectively used by plants for growth and development to the total water applied during irrigation.

6. **What are the essential nutrients for plant growth?**

   Essential nutrients include macronutrients (nitrogen, phosphorus, potassium) and micronutrients (iron, zinc, manganese etc.).

7. **How does the soil's water-holding capacity influence irrigation practices?**
   Soil with high water-holding capacity requires less frequent irrigation, while soil with low capacity needs more frequent watering. This determines irrigation schedules and techniques.

8. **Write a short note on the Temperate fruit zone.**
   The temperate fruit zone includes regions with cold winters and moderate summers. Common fruits grown are apples, pears, and cherries. These crops require chilling hours for proper flowering and fruiting.

9. **How can horticultural therapy be used in healthcare settings to improve physical and mental well-being?**
   Gardening activities reduces stress, enhance emotional well-being of patients with mental health issues or disabilities.

10. **What are Soil Moisture Sensors?**

    The devices used to measure the water content in the soil.

11. **What is air layering?**
    It is an artificial mode of plant propagation.

12. **Write about the nutrient source of an aquaponics system.**
    In aquaponics, fish waste acts as the primary nutrient source for plants.

(10x1=10)

Part B: Short Essay Type Questions

Answer any six questions. Each question carries 5 marks.

13. **Describe the process of creating a garden layout plan.**
    Creating a garden layout plan involves:
    - Find out proper site
    - Assessing the site for sunlight, soil and drainage conditions.
    - Dividing the space into different zones (e.g., lawn, flowerbeds).
    - Selecting suitable plants.
    - Incorporating pathways, irrigation systems, and decorative elements.

14. **Evaluate the effectiveness of polyhouses in extending the growing season for various crops.**
    Poly houses provide controlled environments that protect crops from extreme weather and pests, extend growing seasons, and enable off-season production. They also enhance yield and quality but require high initial investment and maintenance.

15. **How does evapotranspiration influence the need for irrigation?**

    Evapotranspiration, the combined loss of water through evaporation and transpiration, determines the water requirements of plants. High evapotranspiration rates in hot, dry climates increase irrigation frequency to maintain soil moisture. **(Liberal valuation)**

16. **How does soil nutrient management help attain sustainable agrarian culture?**

    Proper nutrient management, such as using organic fertilizers and crop rotation, prevents soil degradation, maintains fertility, and minimizes environmental pollution, supporting long-term agricultural productivity.

17. **How does post-harvest technology preserve the quality and shelf life of horticultural crops?**

    Post-harvest technology includes proper handling, cooling, storage, and packaging methods to reduce spoilage, retain nutritional value, and ensure marketability. Techniques like cold storage and modified atmosphere packaging are commonly used. **(Liberal valuation)**

18. **What are the challenges faced by the horticulture industry in developing countries?**

    Challenges include

    - Limited access to advanced technology,
    - Inadequate storage and transport facilities,
    - Lack of skilled labour,
    - Water scarcity, and
    - Vulnerability to climate change.
    - Proper support from government (Consider any suitable 5 points)

19. **How do you prepare seeds for planting?**

    Seed preparation involves:

    - Selecting viable seeds.
    - Treating seeds with fungicides or insecticides.
    - Soaking or scarifying seeds to enhance germination.
    - Sowing seeds at the appropriate depth and spacing.

20. Assess the environmental benefits of aeroponics over traditional farming.

Aeroponics reduces water usage by 90%, eliminates the need for soil, minimizes pesticide use, and promotes faster growth with fewer resources. It also supports vertical farming, conserving land and reducing deforestation.

(6x5=30)

Part C: Essay Type Questions

Answer any one question. Question carries 10 marks.

21. **Create a plan for implementing AI-driven precision agriculture in a small-scale farm.**

    - **Introduction:** Importance of AI in optimizing farm productivity.
    - **Components:** Use of drones for monitoring, sensors for soil health, and AI tools for data analysis.
    - **Implementation Steps:**
        1. Install IoT-based soil and weather sensors.
        2. Use drones to collect data on crop health.
        3. Employ AI algorithms to predict pest attacks and irrigation needs.
    - **Benefits:** Improved resource efficiency, reduced costs, and enhanced yield. **(Liberal valuation)**

22. **Discuss in detail the structure, functions, and applications of greenhouses in agriculture and horticulture.**

    - **Structure:** Framework covered with transparent material, ventilation systems, and temperature controls.
    - **Functions:** Protects plants from adverse weather, controls temperature and humidity, and extends growing seasons.
    - **Applications:** Growing high-value crops, seedling production, and research on plant varieties.

(1x10=10)